

Truth Table/Characteristics table/Excitation Table

J. K FF

Truth Table/TT

clk	J	K	Q_{n+1}
0	X	X	Q_n } memory
1	0	0	Q_n }
1	0	1	0 $\leftarrow$ Reset
1	1	0	1 $\leftarrow$ set
1	1	1	$\overline{Q_n} \leftarrow$ Toggle

Characteristics and Excitation tables are very useful to design counters. It is required to convert ff. Ch. Table

The next state of JK flip-flop is dependent on the input i.e. J & K along with the previous state which is Q_n . So, the inputs are Q_n J K. And the output is Q_{n+1}

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Characteristics table

now, we need to deduce the excitation table based on the characteristic table. The input of the excitation table is present state and next state Q_n , Q_{n+1} , and the output is J K.

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

You can Remember like for J

0 $\rightarrow$ 1 $\rightarrow$ don't care $\rightarrow$ don't care
For K

don't care $\rightarrow$ don't care $\rightarrow$ 1 $\rightarrow$ 0

- lets find $Q_n = 0$ & $Q_{n+1} = 0$; We find 2 conditions where J is 0 for both & K gets changed. So, we take J is 0 & K is don't care. the value is only dependent on J. not K. thats why we took ~~set~~ K don't care.
- lets find $Q_n = 0$, $Q_{n+1} = 1$; here J=1 but K changes so, J=1, K=don't care
- lets, find $Q_n = 1$, $Q_{n+1} = 0$; here K=1; J=dc
- lets, find $Q_n = 1$, $Q_{n+1} = 1$; here K=0; J=dc

2 bit Synchronous Counter Using JK Flip-flops

Number of FF: 2

① Excitation Table for JK FF.

③ State Diagram = how many states = $2^2 = 4$

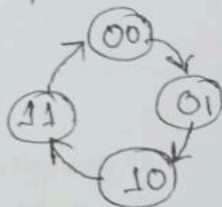
④ Truth Table for present state and Next state

⑤ K-Map to derive equation

⑥ Logical Diagram

③ state diagram

Q_1	Q_0
0	0
0	1
1	0
1	1



② Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

④ Truth Table Using state Diagram

Present state	Next state	J, K ₁	J ₀ , K ₀
Q_1 Q_0	Q_1^* Q_0^*		
0 0	0 1	0 X	1 X
0 1	1 0	1 X	X 1
1 0	1 1	X 0	1 X
1 1	0 0	X 1	X 1

⑤ Now K-Map for J_1 , K_1

Boolean expression.

J_1

Q_1	Q_0	
0	0	0
0	1	1
1	0	1
1	1	1

$$J_1 = Q_0$$

K_1

Q_1	Q_0	
0	0	0
0	1	X
1	0	X
1	1	1

$$K_1 = Q_0$$

Now K-map for J_0 , K_0

J_0

Q_1	Q_0	
0	0	1
0	1	1
1	0	X
1	1	X

$$J_0 = 1$$

K_0

Q_1	Q_0	
0	0	0
0	1	X
1	0	X
1	1	1

$$K_0 = 1$$

if we consider X to be 1 we can see that all cells have value 1, so the expression will have 1,

⑥ So, Now we have the boolean expressions of the inputs. We will draw the ckt using Expression.

$$J_0 = Q_0, K_1 = Q_0, J_0 = 1, K_0 = 1$$

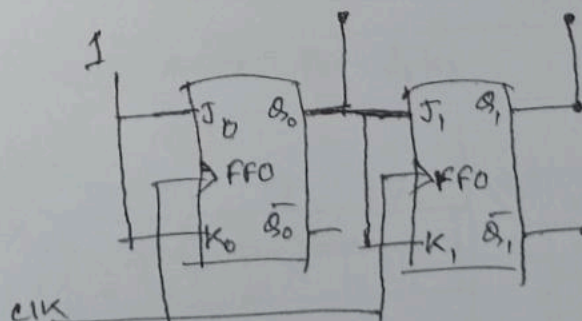


Figure: 2 bit synchronous Counter (up) using JK Flip-Flops.

Designing Counters Using Methods (3 bit)

Example: 3 bit synchronus Up counter Using JK FF.

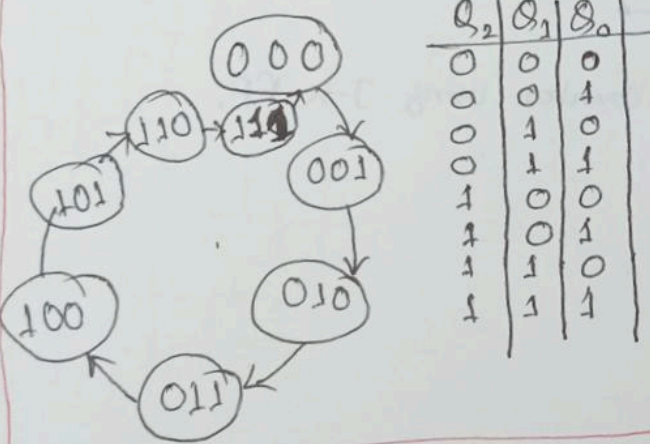
- 1) Number of FF used: since we need to design 3 bit counter we need 3 FFs
- 2) Excitation table for JK ff
- 3) State Diagram = how many states = $2^n = 2^3 = 8$
- 4) Truth Table for present state and next state
- 5) K-Map $\rightarrow$ derive equation
- 6) Logical Diagram.

Excitation Table for JK ff

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

State diagram

since we are designing 3 bit up counter we will have 8 states (2^n)



Truth Table Using State Diagram

Present state			Next state			MSB		LSB				
Q_2	Q_1	Q_0	Q_2^*	Q_1^*	Q_0^*	J_2	K_2	J_1	K_1	J_0	K_0	
0	0	0	0	0	1	0	X	0	X	1	X	0
0	0	1	0	1	0	0	X	1	X	X	1	1
0	1	0	0	1	1	0	X	X	0	1	X	2
0	1	1	1	0	0	1	X	X	1	X	1	3
1	0	0	1	0	1	X	0	0	X	1	X	4
1	0	1	1	1	0	X	0	1	X	X	1	5
1	1	0	1	1	1	X	0	X	0	1	X	6
1	1	1	0	0	0	X	1	X	1	X	1	7

K-Map Using the Truth Table

First let us construct TT for $J_2 K_2$

$Q_2 Q_1$	00	01	11	10
Q_0	0	2	6 X	4 X
1	1	3 1	7 X	5 X

$$J_2 = Q_0 Q_1$$

K_2

$Q_2 Q_1$	00	01	11	10
Q_0	0 X	2 X	6 X	4
1	1 X	3 X	7 1	5

$$K_2 = Q_0 Q_1$$

Now, Next K-Map for $J_1 K_1$

$Q_2 Q_1$	00	01	11	10
Q_0	0	2 X	6 X	4
1	1 1	3 X	7 X	5 1

$$J_1 = Q_0$$

$Q_2 Q_1$	00	01	11	10
Q_0	0 X	2	6	4 X
1	1 X	3 1	7 1	5 X

$$K_1 = Q_0$$

Now, Next K-Map for $J_0 K_0$

$Q_2 Q_1$	00	01	11	10
Q_0	0 1	2 1	6 1	4 1
1	1 X	3 X	7 X	5 X

if all the cell has 1 in a group then you can take the expression as $J_0 = 1$

$Q_2 Q_1$	00	01	11	10
Q_0	0 X	2 X	6 X	4 X
1	1 1	3 1	7 1	5 1

$$K_0 = 1$$

Now we have all our equations in our hand.

$$J_2 = Q_0 Q_1, K_2 = Q_0 Q_1, J_1 = Q_0, K_1 = Q_0, J_0 = 1, K_0 = 1$$

⑥ Now we need to draw the circuit using the equations.
First draw the FFs. connect all clocks together for a single clock.

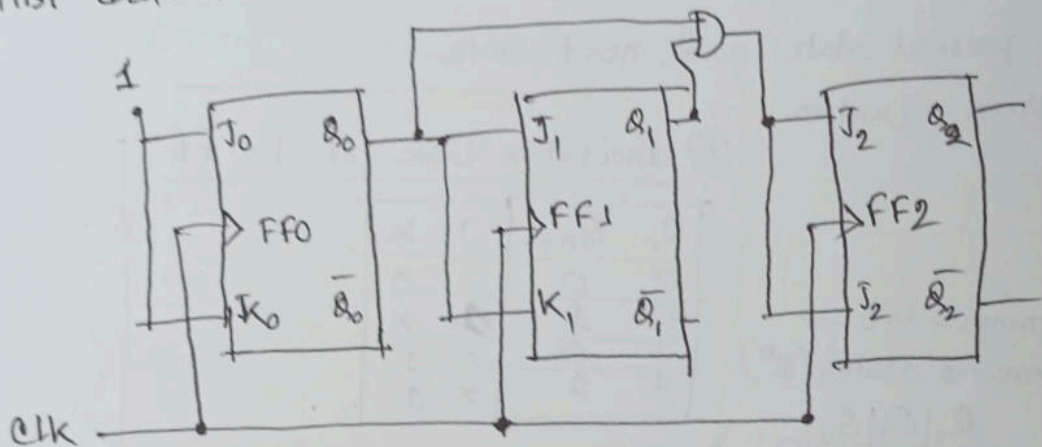


Figure: 3 bit synchronous Up-Counter using J-K FF.

Designing 4 bit synchronous Binary counter

①

No. of state = 16 because 4 bit ($2^n = 2^4 = 16$)

range of counting (0 to $2^n - 1$) $\rightarrow 0$ to $2^4 - 1 = 0$ to 15

How many FF. $n=4$, So, Number of FF is 4.

lets start with the state diagram.

No of FF = No. of bits
for mod

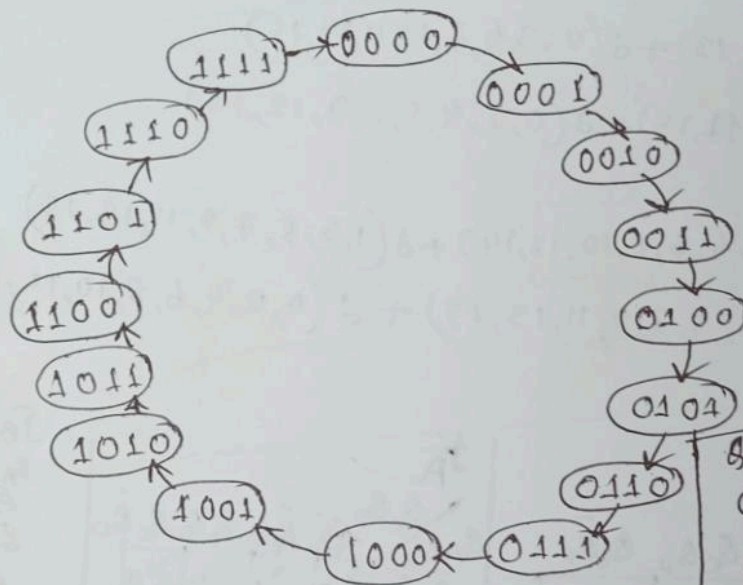
$$N \leq 2^n$$

$$16 \leq 2^4 = 16 = 16$$

	Q_A	Q_B	Q_C	Q_D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

A B C D
↑ ↑
MSB LSB

State Diagram



Now lets make the state table

Ex table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Present state				Next state				Input of FF							
Q_A	Q_B	Q_C	Q_D	Q_{A+1}	Q_{B+1}	Q_{C+1}	Q_{D+1}	J_A	K_A	J_B	K_B	J_C	K_C	J_D	K_D
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	1	X
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	0	X	X	1	0	X	1	X
1	0	0	1	1	0	1	0	0	X	X	0	1	X	1	X
1	0	1	0	1	0	1	1	0	X	X	0	X	0	1	X
1	0	1	1	1	1	0	0	0	X	1	X	X	1	X	1
1	1	0	0	1	1	0	1	0	X	X	0	0	X	1	X
1	1	0	1	1	1	1	0	0	X	X	0	1	X	X	1
1	1	1	0	0	0	0	0	1	X	X	0	X	0	1	X
1	1	1	1	0	0	0	0	0	X	X	0	X	0	1	X
1	1	1	1	0	0	0	0	0	X	X	0	X	0	1	X

Now, constructing All eight inputs eqn.

$$J_A = \sum m(7) + d(8, 9, 10, 11, 12, 13, 14, 15)$$

$$K_A = \sum m(15) + d(0, 1, 2, 3, 4, 5, 6, 7)$$

$$J_B = \sum m(3, 11) + d(4, 5, 6, 7, 12, 13, 14, 15)$$

$$J_B = \sum m(7, 15) + d(0, 1, 2, 3, 8, 9, 10, 11)$$

$$J_C = \sum m(1, 5, 9, 13) + d(2, 3, 6, 7, 10, 11, 14, 15)$$

$$K_C = \sum m(3, 7, 11, 15) + d(0, 1, 4, 5, 8, 9, 12, 13)$$

$$J_D = \sum m(0, 2, 4, 6, 8, 10, 12, 14) + d(1, 3, 5, 7, 9, 11, 13, 15)$$

$$K_D = \sum m(1, 3, 5, 7, 9, 11, 13, 15) + d(0, 2, 4, 6, 8, 10, 12, 14)$$

J_A

	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
$A\bar{B}$	X ₁₂	X ₁₃	X ₁₅	X ₁₄
AB	X ₈	X ₉	X ₁₁	X ₁₀

$J_A = \bar{B}\bar{C}D$

J_A

$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$	
$\bar{A}\bar{B}$	X ₀	X ₁	X ₃	X ₂
$\bar{A}B$	X ₄	X ₅	X ₇	X ₆
$A\bar{B}$	X ₁₂	X ₁₃	X ₁₅	X ₁₄
AB	X ₈	X ₉	X ₁₁	X ₁₀

$K_A = \bar{B}\bar{C}D$

$$J_A = K_A = \bar{B}\bar{C}D$$

$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
$\bar{A}\bar{B}$	0	1	3
$\bar{A}B$	X ₄	X ₅	X ₇
$A\bar{B}$	X ₁₂	X ₁₃	X ₁₅
AB	X ₈	X ₉	X ₁₁

$J_B = \bar{B}\bar{C}D$

K_B

	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
$\bar{A}\bar{B}$	X_0	X_1	X_3	X_2
$\bar{A}B$	4	5	7	6
$A\bar{B}$	12	13	15	14
AB	X_8	X_9	X_{11}	X_{10}

J_C

	$\bar{B}_C \bar{D}$	$\bar{B}_C D$	$B_C \bar{D}$	$B_C D$
$\bar{A}_B \bar{A}_B$	0	1	X ₃	X ₂
$\bar{A}_B A_B$	4	5	X ₇	X ₆
$A_B \bar{A}_B$	12	13	X ₁₅	X ₁₄
$A_B A_B$	8	9	X ₁₁	X ₁₀

J_C

K_C

$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
$\bar{A}\bar{B}$	X ₀	X ₁	13
$\bar{A}B$	X ₄	X ₅	17
$A\bar{B}$	X ₁₂	X ₁₃	15
AB	X ₈	X ₉	11

$K_C = \bar{B}\bar{C}D$

$$J_C = K_C = \bar{B}\bar{C}D$$

J_D

$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$	
$\bar{A}\bar{B}$	1	X ₁	X ₃	2
$\bar{A}B$	4	X ₅	X ₇	6
$A\bar{B}$	12	X ₁₃	X ₁₅	14
AB	1	X ₉	X ₁₁	10

K_D

$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$	
$\bar{A}\bar{B}$	X ₀	1	13	X ₂
$\bar{A}B$	X ₄	15	7	X ₆
$A\bar{B}$	X ₁₂	1	15	X ₁₄
AB	X ₈	1	11	X ₁₀

$$J_D = K_D = \bar{B}\bar{C}D$$

Now we have all our eqn in our hand

②

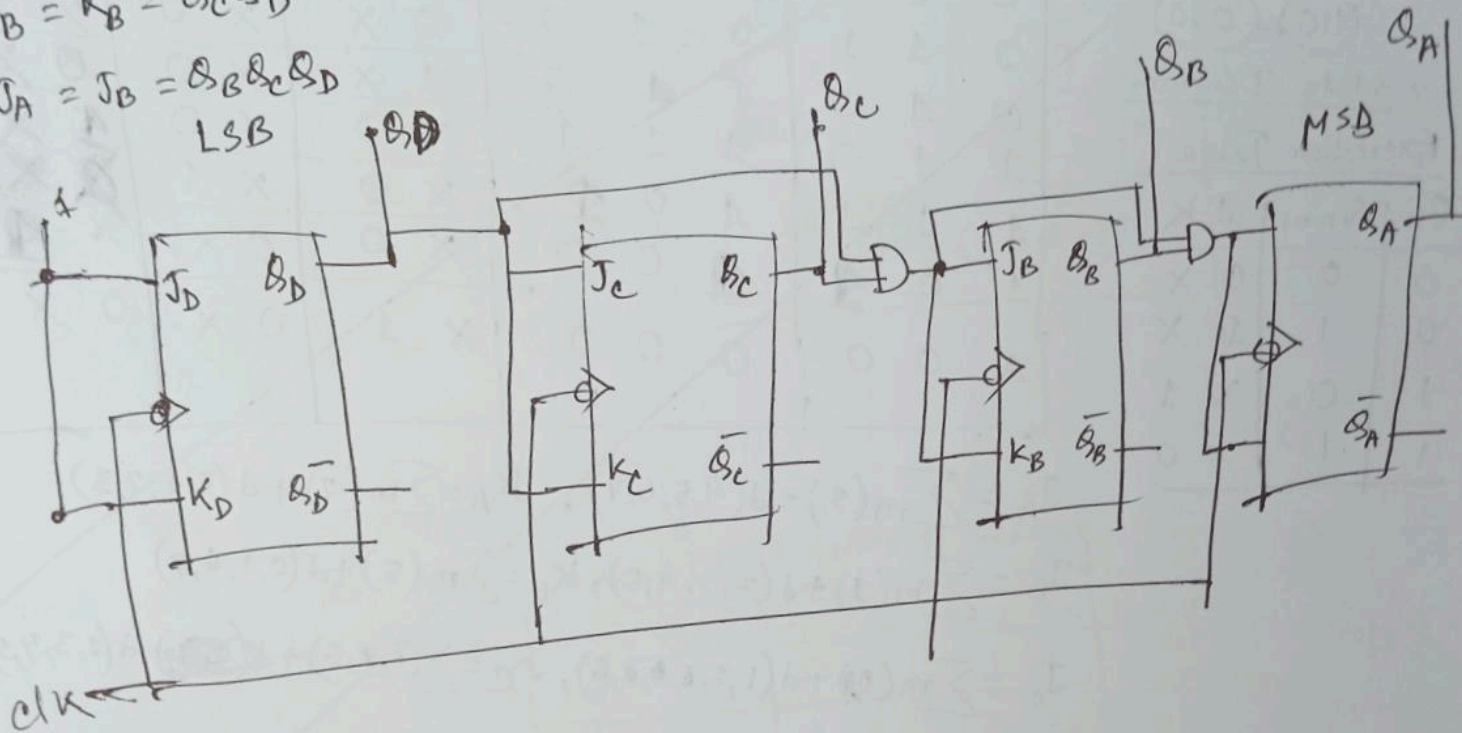
$$J_D = K_D = 1$$

$$J_C = K_C = Q_D$$

$$J_B = K_B = Q_C Q_D$$

$$J_A = J_B = Q_B Q_C Q_D$$

LSB



non
cha
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You
0 → 1
Fork
don'tc

Designing a 3 bit Gray Code counter

(3)

GRAY CODE TT

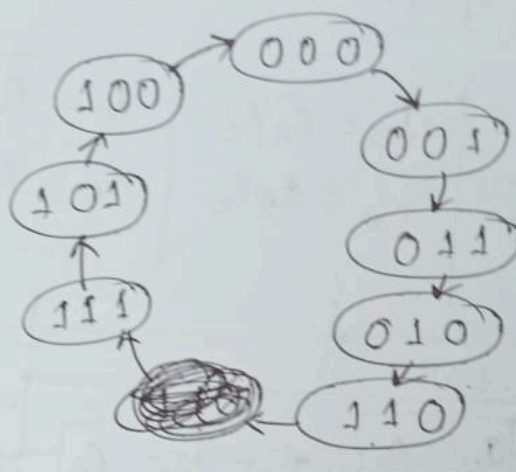
Use Mirror Image Technique to make the truth table

3 bit Gray

	G_2	G_1	G_0
0	0	0	0
1	0	0	1
2	0	1	1
3	0	1	0
4	1	1	0
5	1	1	1
6	1	0	1
7	1	0	0

Start with one zero and one 1
Mirror then 2 0s and 2 1s
Mirror then 4 0s and 4 1s

State diagram



J K FF Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

State Table preparation

Present state	Next state	Input of FFA	Input of FFB	Input of FFC
Q_A Q_B Q_C	Q_{A+1} Q_{B+1} Q_{C+1}	J_A K_A	J_B K_B	J_C K_C
0 0 0	0 0 1	0 X	0 X	1 X
0 0 1	0 1 1	0 X	1 X	X 0
0 1 1	0 1 0	0 X	X 0	X 1
0 1 0	1 1 0	1 X	X 0	0 X
1 1 0	1 1 1	X 0	X 0	1 X
1 1 1	1 0 1	X 0	X 1	X X
1 0 1	1 0 0	X 0	0 X	X 1
1 0 0	0 0 0	X 1	0 X	0 X

$$J_A = \sum m(3) + d(4, 5, 6, 7)$$

$$K_A = \sum m(7) + d(0, 1, 2, 3)$$

$$J_B = \sum m(1) + d(2, 3, 4, 5)$$

$$K_B = \sum m(5) + d(0, 1, 6, 7)$$

$$J_C = \sum m(0, 4) + d(1, 2, 5, 6)$$

$$K_C = \sum m(2, 6) + d(0, 3, 4, 7)$$

for J A K A

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

(JA)

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

$$J_A = \bar{Q}_B \bar{Q}_C$$

$$K_A = \bar{Q}_B \bar{Q}_C$$

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

(JB)

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

$$J_B = \bar{Q}_A \bar{Q}_C$$

$$K_B = \bar{Q}_A \bar{Q}_C$$

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

(JC)

Q_C	$\bar{Q}_C$	Q_C	$\bar{Q}_C$
0	1	0	1
1	0	1	0
0	1	1	0
1	0	0	1

$$J_C = \bar{Q}_A \bar{Q}_B + \bar{Q}_A \bar{Q}_B$$

$$= \bar{Q}_A \bar{Q}_B + \bar{Q}_A \bar{Q}_B$$

$$K_C = \bar{Q}_A \bar{Q}_B + \bar{Q}_A \bar{Q}_B$$

$$= \bar{Q}_A \bar{Q}_B$$

Now we have All the eqns at our hand.

The Number of FFs is 3.

eqns are

$$J_A = Q_B \bar{Q}_C$$

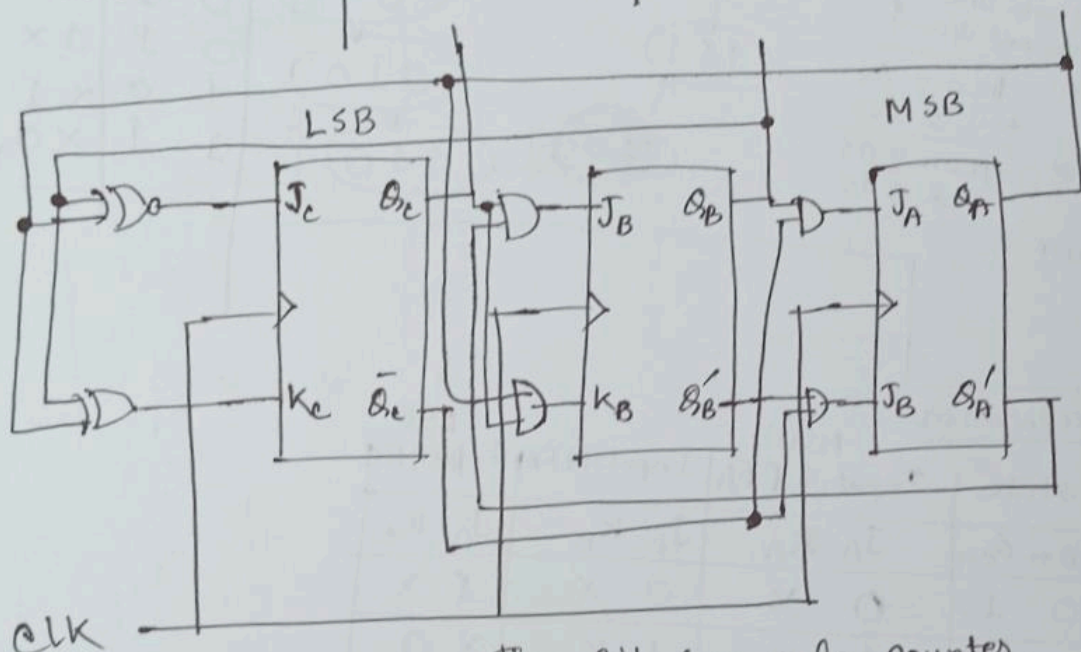
$$K_A = \bar{Q}_B \bar{Q}_C$$

$$J_B = \bar{Q}_A Q_C$$

$$K_B = Q_A Q_C$$

$$J_C = \bar{Q}_A \oplus Q_B$$

$$K_C = Q_A \oplus Q_B$$



clk

Three Bit Gray code counter